

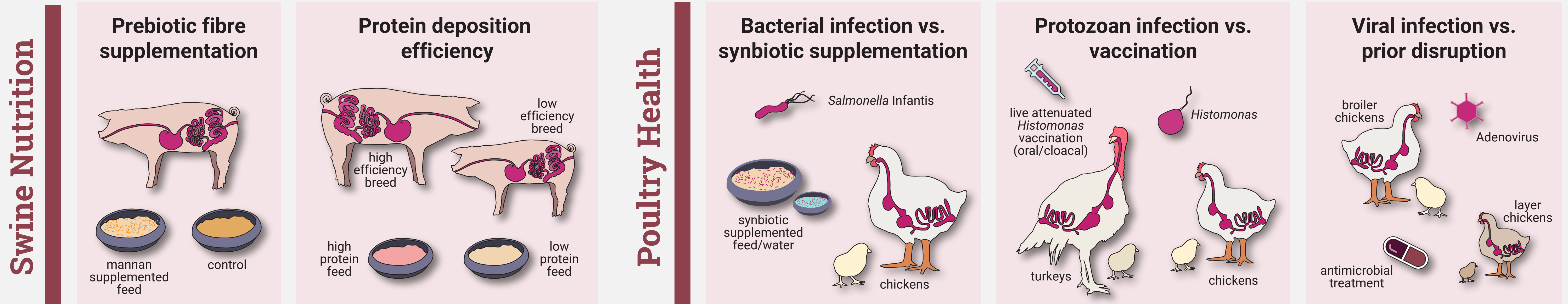
Zooming into Livestock Microbiomes via Micro-Scale Spatial Metagenomics



UNIVERSITY OF COPENHAGEN

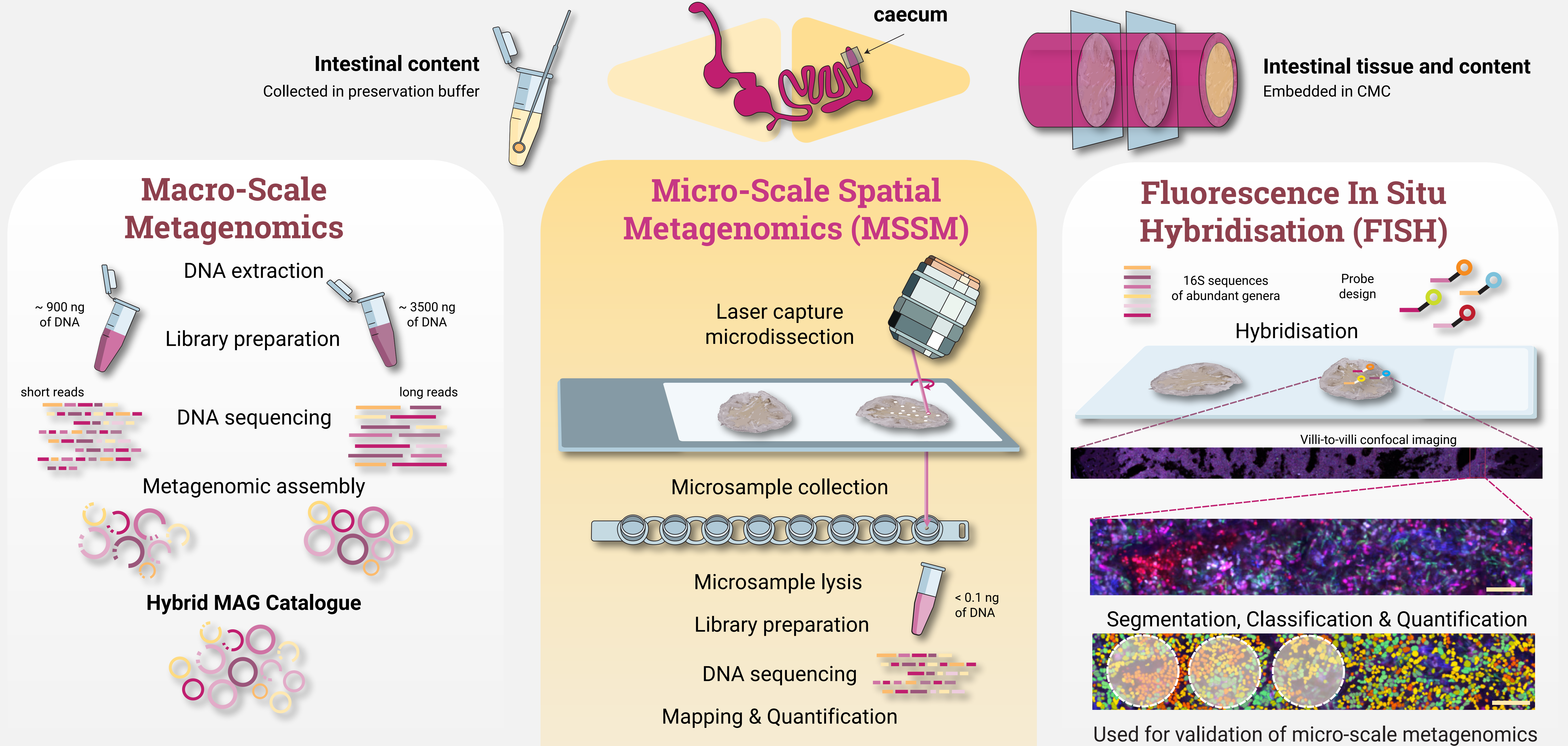
INTRODUCTION

Understanding the interplay between animals and their associated microorganisms is essential for improving animal health, welfare, and production. In 3D'omics, we generate multi-omic data to unveil microbe-microbe and animal-microbe interactions that have so far remained undetected.



METHOD DEVELOPMENT

Conventional samples were collected for metagenomics, metatranscriptomics, metabolomics, and host genomics & transcriptomics analyses. Snap-frozen tissue was collected for targeted, zoomed-in, spatial analyses: micro-metagenomics, microscopy, micro-metabolomics.

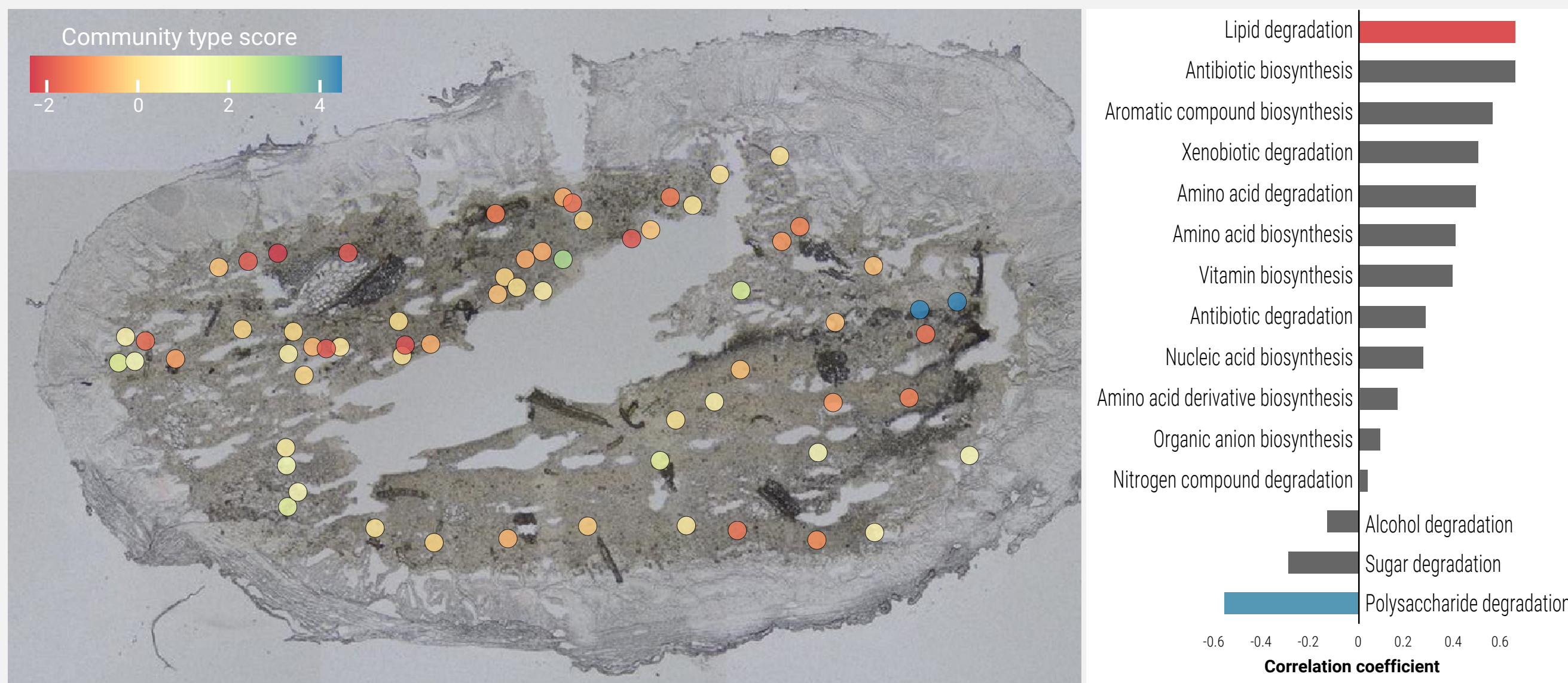


APPLICATION

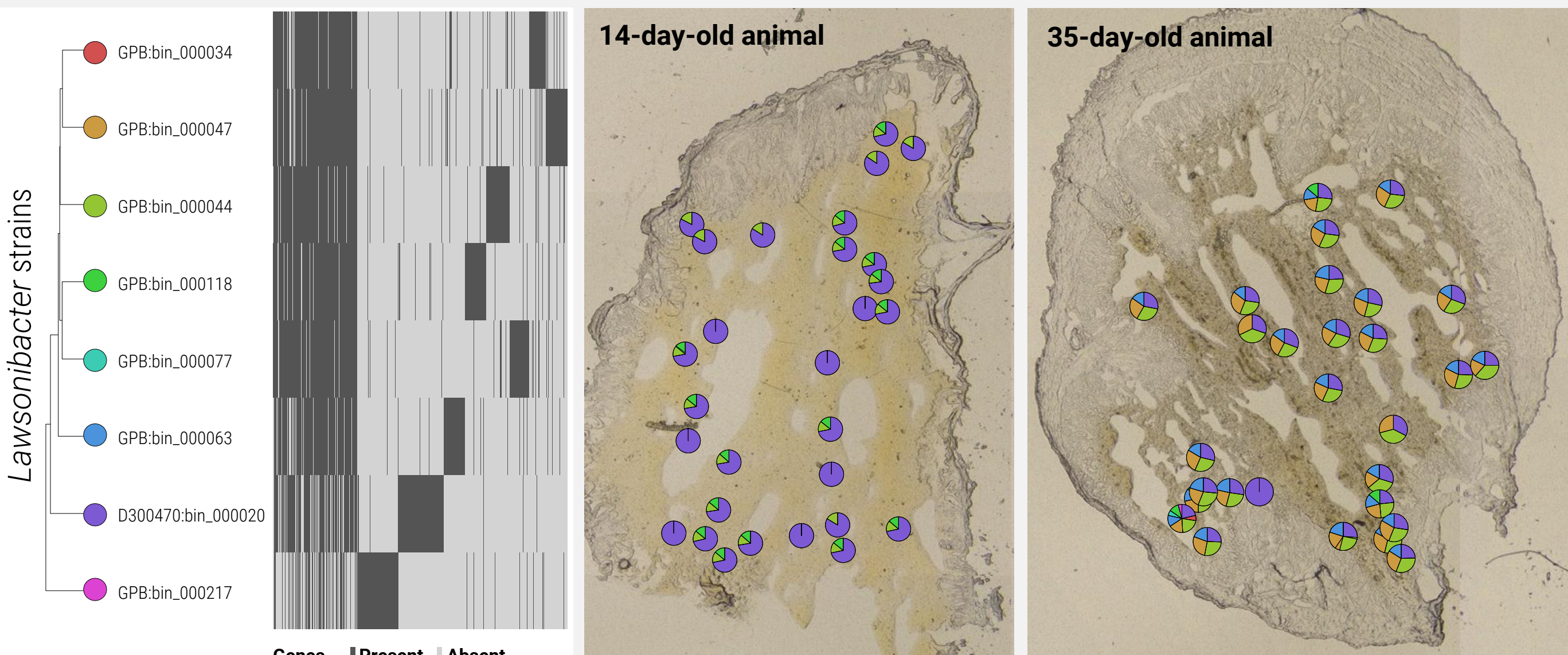
We generated macro-metagenomic data from >500 animals. From 44 animals, we generated > 5.500 micro-samples.

We are currently exploring these micron-scale bacterial networks to see precisely how different treatments affect bacterial interactions, relationships that were up to now obscured by conventional bulk sampling.

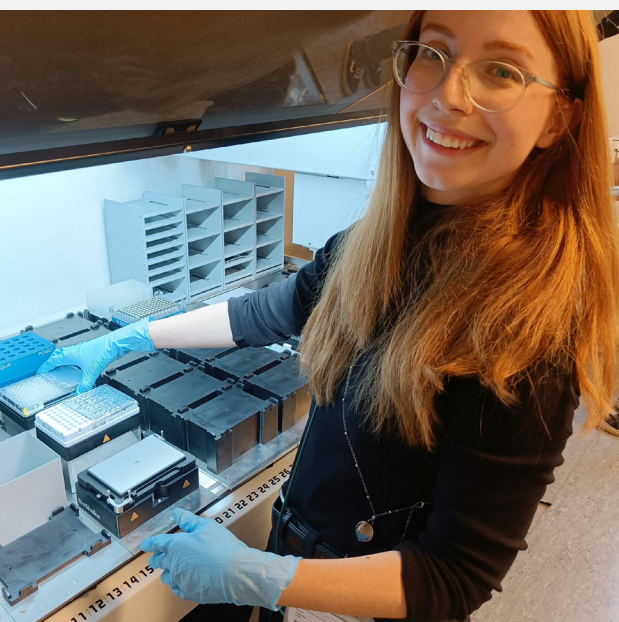
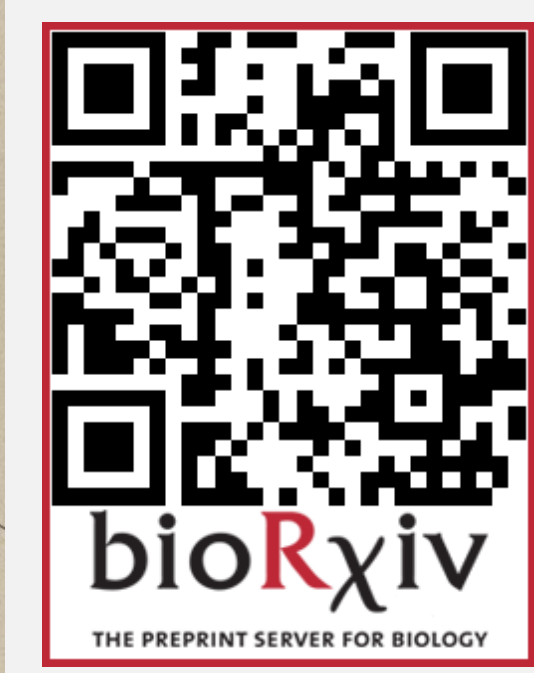
Combined spatial, phylogenetic, and functional analyses of the microsamples may detect spatial signals, if the digesta is organised as such. ▼



Strain level variations can be detected across the cryosections, and across different animals. ▼



Find our methods pre-print here:



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3D'omics

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